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Hotel Reservation Management System

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Abstract

Traditional hotel reservation processes often suffer from booking errors, limited availability visibility, and high manual workload, which negatively affect guest experience and operational efficiency. To address these challenges, this project develops a Hotel Reservation Management System (HRMS) that automates reservation procedures and centralizes hotel operations within a single digital platform. The system allows guests to search rooms, book online, complete secure payments, and redeem reward points, while enabling administrators and staff to manage room availability, inventory, promotions, reports, and housekeeping tasks. The system is designed using a layered architecture to ensure security, scalability, and structured information flow, with HTML, CSS, and JavaScript forming the user interface, and PHP and MySQL supporting server-side logic and database management. A functional prototype was implemented to demonstrate real-time interactions, workflow automation, and role-based system access. The HRMS is expected to reduce booking conflicts, improve accuracy, enhance user experience, and support better decision-making in hotel environments.

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1. Introduction

The rapid digital transformation in the hospitality sector has increased the need for reliable online reservation platforms that enhance operational efficiency and improve guest satisfaction. Modern hotel reservation solutions are expected to provide real-time room availability, secure online booking, accurate inventory control, and seamless synchronization between hotel websites and external booking channels. These capabilities help prevent double-booking, reduce manual workload, and ensure consistent booking experience for guests.

This project proposes the design and development of a Hotel Reservation Management System (HRMS) tailored for small to mid-sized hotels. The system aims to streamline core booking activities while supporting the full end-to-end reservation process. Guests can search available rooms, compare options by price and features, complete secure online reservations, and receive immediate confirmation. They can also modify or cancel bookings through the system without delay, while room availability remains accurate and always synchronized.

Inspired by reservation and channel management capabilities, HRMS integrates secure payment handling, centralized control, and real-time data updates into a simplified solution suitable for smaller hotels. This project demonstrates how a well-designed reservation system can automate routine hotel tasks, minimize staff errors, and enhance satisfaction for both employees and guests.

2. Methodology Description

2.1 Requirement Gathering and Analysis

It focuses on identifying what the Hotel Reservation Management System must achieve for both users and the hotel's operational processes. This stage defines the functional, non-functional, and user requirements needed to support booking, room management, staff operations, and guest interactions. Requirements were collected by analyzing the hotel's existing process, identifying problems such as dependence on third-party booking portals, profit loss due to commission fees, and the need for a unified system for booking, housekeeping, reporting, and inventory management. Interviews and observations with hotel staff and administrators guided requirement identification.

2.2 System Design and Architecture

The study [3] highlights the importance of selecting modern architectural patterns that enhance scalability, security, and maintainability. After gathering system requirements, a structured design approach was used to translate these requirements into a clear architecture model. During this phase, the main system components were identified, their relationships were outlined, and a layered architecture was selected to organize the system into separate functional layers. This approach supports maintainability, security, and future expansion. A detailed description of the four-tier architecture is provided in Section 4.

2.3 Technology selection and implementation

The technologies selected to develop the Hotel Reservation Management System will be such that they support a secure, scalable, and interactive environment that can handle a wide range of hotel operations, including room availability updates, multi-role user interactions, creation and modification of reservations, payment handling, promotions, reward points, inventory updates, staff workflows, and cleaning task management. The selected technology stack ensures reliability, maintainability, and smooth performance throughout the entire system.

2.3.1 Front-End Technologies

HTML5, CSS3, and JavaScript were used in the development of a responsive and user-friendly interface presented in the Figma prototype. These technologies have allowed the system to display:

- An interactive dashboard displaying KPIs on total rooms, occupancy rate, active bookings, and revenue.
- Room listings, detailed room descriptions, and availability of rooms.
- Reservation and payment pages for guests.
- Staff pages for check-in/out, for room status updates, inventory updates, and cleaning notifications.
- Admin pages for managing promotions and rewards and generating daily reports.

JavaScript helps in enhancing users' experience with its live form validation, interactively changing charts, room statuses that dynamically update, and easy navigation across various devices. CSS3 provides a modern visual design and appearance, while maintaining consistency with the prototype layout.

2.3.2 Back-End Technology

The application logic is implemented using server-side language like PHP since it goes well with MySQL and finds extensive use in hotel booking platforms. The backend handles the following:

- Online reservation processing and modification
- Price calculation and date validation
- Real-time updates to room availability
- User and staff authentication
- Management of reward points and voucher codes
- Inventory update and cleaning task workflows
- Promotions by administrators

- Notification Handling
- Daily report generation

PHP has stability and security features, as well as powerful database integration, which is appropriate for managing the full booking lifecycle and supporting all hotel roles described in the system.

2.3.3 Database Technology

MySQL is being used as the central relational database for the HRMS. Based on the system requirements outlined in the project report, the database stores the following information:

- Guest accounts and staff profiles
- Room types, pricing, and availability
- Reservation details and modification logs
- Payment transactions
- Promotions, reward points, and voucher codes
- Inventory levels for each room
- Cleaning schedules and housekeeping records

MySQL provides data integrity, high performance for queries, and real-time availability, all very critical in this scenario for avoiding double-bookings and maintaining synchronized hotel operations.

2.3.4 Security Measures

Since the HRMS will handle very sensitive information, including login credentials, payment information, and personal reservation details, multiple mechanisms for security are implemented:

- HTTPS encryption to secure all client–server communication
- Hashing of passwords for secure authentication

- RBAC, allowing each user to access only allowed features: guest, admin, staff, cleaning staff
- Session management ensures unauthorized access is prevented
- Input validation, to mitigate SQL injection and automated bot bookings
- Secure payment workflow, following the process outlined in the activity diagram

These measures together protect system data, prevent abuse, and ensure reliable booking operation processes.

2.4 Web/Desktop Application Development

Building on the layered design described above, HRMS was developed as an interactive, transaction-based web application intended for desktop use through a standard browser. The system relies on a centralized MySQL database that updates instantly to synchronize hotel operations. Any user action, such as searching for rooms, confirming a reservation, modifying or canceling a booking, applying promotions, updating room availability, viewing reports, managing inventory, or updating room-cleaning status, directly updates the database so changes appear immediately for all authorized roles. The Presentation Layer includes role-specific interfaces for guests, administrators, front desk staff, staff members, and cleaning staff. The Application/Business Logic Layer executes the core workflows described in the use cases and sequence diagrams, including availability checks, pricing with discounts, payment processing, reservation status updates, inventory control, rewards tracking, and housekeeping coordination. The Data Layer stores and retrieves system entities (rooms, reservations, users, payments, promotions, inventory items, and cleaning records) while ensuring integrity and preventing double-booking. Finally, the Security Layer protects the system through authentication, password hashing, secure sessions, RBAC, input validation, HTTPS communication, and secure payment handling. Overall, HRMS supports essential hotel workflows in an integrated manner. Guests can complete their booking journey electronically, administrators can monitor performance and manage operations, front desk staff can handle arrivals and departures, staff can update inventory and availability, and cleaning teams can update room readiness. This

development structure aligns with the finalized functional and non-functional requirements, ensuring reliability, security, and ease of use.

3. Requirements

3.1 User Requirements

- **All Users**
  - Enter user ID/email and password to login.
  - Logout.
- **Guest**
  - Register a new account.
  - View room availability.
  - Reserve rooms with selected dates and room count.
  - Payment to complete reservation.
  - Exchange voucher codes using reward points during payments for the next reservation.
  - Receive reservation confirmation notifications.
- **Administrator**
  - Register new staff accounts.
  - Generate and view daily reports.
  - Set hotel promotions.
- **Front Desk Staff**
  - Confirm/cancel reservations.
  - Check guests in/out.
  - Change rooms when necessary.
- **Staff**
  - Update room availability.
  - Update inventory.
- **Cleaning Staff**
  - Receive cleaning notifications when guests check out.
  - Update cleaning status.

3.2 System Requirements

- **Hardware Requirements**
  - CPU.
  - RAM.
  - Storage.
- **Software Requirements**
  - Languages: HTML, CSS, PHP, JavaScript
  - Design Tool: Draw.io, Figma
  - Database: MySQL
  - Operating System: Windows 11

3.3 Functional Requirements

- **Registration Module**
  - The system must allow new guests to register before logging in.
  - Admin can register new staff accounts.
  - Admin account registration is not permitted.
- **Login / Logout Module**
  - Guests login using email + password.
  - Admin and staff login using user ID + password.
  - System must validate credentials and show “login failed” when invalid.
  - Successful login redirects to the main page.
  - Users can log out from their account at any time.
- **Reservation Module**
  - Display room availability based on selected check-in/check-out dates.
  - Allow guests to reserve rooms.
  - Guests complete payment to confirm the reservation.
  - Front desk staff can confirm/cancel reservations and send confirmation emails.
  - Front desk staff can change rooms if needed.
  - Front desk staff can check guests in/out.
- **Report Module**
  - Admin can generate and view daily hotel activity reports.

- **Management Module**
  - Admin can set and display promotions.
  - Staff can update room availability.
  - System sends notifications to cleaning staff when rooms need cleaning.
  - Cleaning staff can update room cleaning status.
- **Inventory Module**
  - Staff can update inventory for each room.
- **Reward Module**
  - Guests can redeem voucher codes using reward points.
  - Admin manages the reward options.

3.4 Non-Functional Requirements

- **Security:** Encrypted access to prevent unauthorized use and bot bookings.
- **Reliability:** Accurate room status to reduce reservation conflicts.
- **Performance:** System should respond within an acceptable time.
- **Usability:** Simple, user-friendly interface.
- **Availability:** Accessible at all times.
- **Operational:** Works on web browsers and easy to maintain.

4. Architecture Design

The architectural design of the Hotel Reservation Management System is a critical link between system requirements and design. It helps define the system's main structural components and their interrelationships, ensuring a comprehensive understanding of how the entire system functions. The reservation management system is designed using layered architecture, organizing the system into layers, each with its own functions and services for the layer above it. This structure ensures protection of sensitive user information, payment and reservation booking details, accurate booking processing, simplified system maintenance, and future scalability. It also supports user roles such as guests, administrators, receptionists, general staff, and cleaning personnel.

1. **Presentation Layer** the layer is everything the user sees and interacts with. Therefore, the layer will provide all user interfaces, including login pages, registration pages, room search pages, room details pages, booking creation pages, payment pages, and pages for confirming, modifying, or canceling bookings, as required by our system. It also includes the interfaces required by staff members such as check-in/check-out pages, room change pages, reservation confirmation pages, daily report pages for administrators, promotion management interfaces, room availability update screens, inventory update screens, and cleaning task pages for cleaning staff. It sends input to the application layer, but only after verifying the user's input, which helps reduce errors.
2. **Application Layer** this layer represents the system's brain, it's where all operations and requests originating from the user interface are processed and returned to. Therefore, this layer handles all the intelligent steps involved, including verifying room availability, calculating prices, processing modifications and cancellations, enforcing system rules, creating reservations, and communicating with the database according to our system requirements. It also manages staff operations, including check-in and check-out, room changes, updating room availability, updating inventory, generating daily reports, handling reward points redemption, and validating voucher codes. This layer then sends the inputs to the data layer to retrieve or update stored information.
3. **Data Layer** this layer represents data storage location and manages hotel operations. This includes user data, staff accounts, room types and prices, reservation records, modification records, and payments, inventory data, promotions, reward point balances, voucher codes, and cleaning task logs, all in accordance with our system requirements. Therefore, it ensures data consistency and prevents double-booking by providing room storage updates after each transaction. This layer also meets non-functional requirements, including reliability (ensuring accurate stored data), availability (ensuring 24/7 data accessibility), and integrity.
4. **Security Layer** this layer represents system protection only. It includes verifying user authenticity, input validity, access permissions, password encryption, payment

data protection, Session management, and attack prevention. Therefore, it ensures data privacy and integrity and prevents unauthorized access. This layer also supports non-functional and system requirements such as user authentication, secure payment processing, and user session management.

So, layered architecture improves separation of concerns, enhances system security, and simplifies future maintenance. It also ensures that the reservation system remains scalable and reliable while handling sensitive user and payment information efficiently.

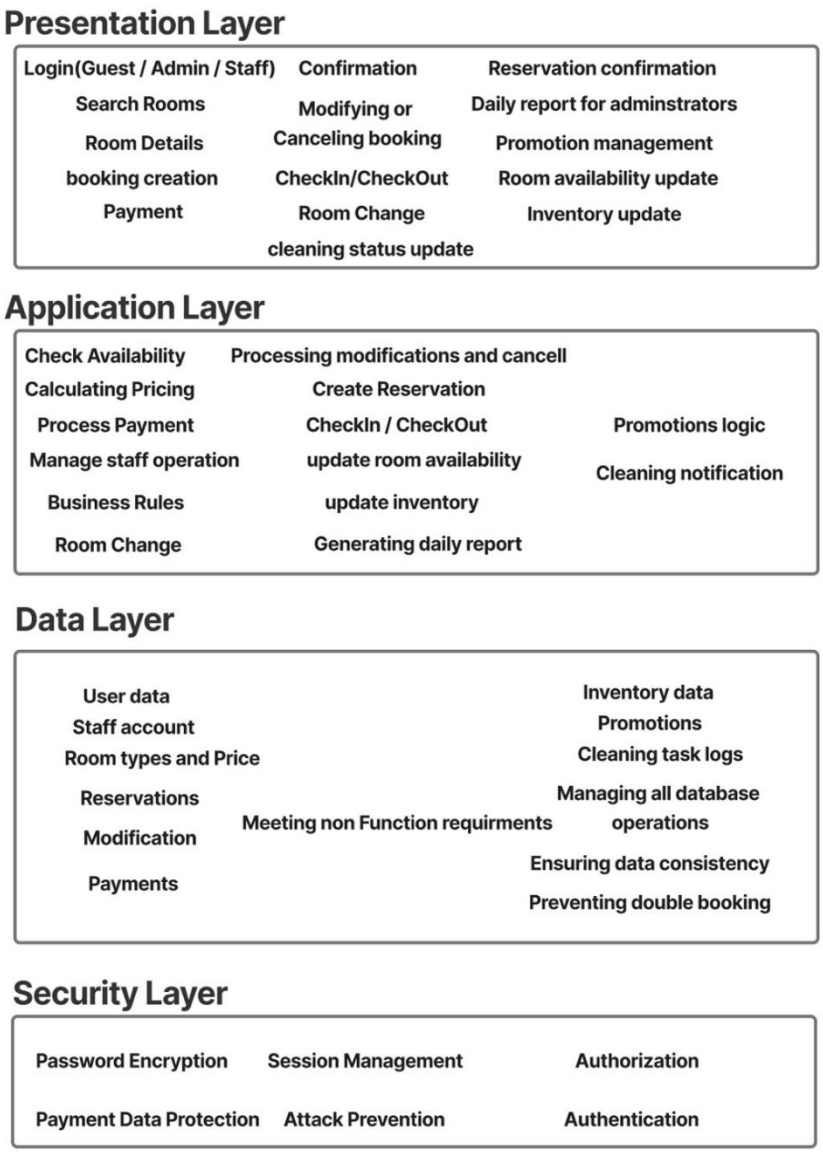

Figure 1: Architecture Design for HRMS.

5. Models

4.1 Use Case Diagrams with Descriptions

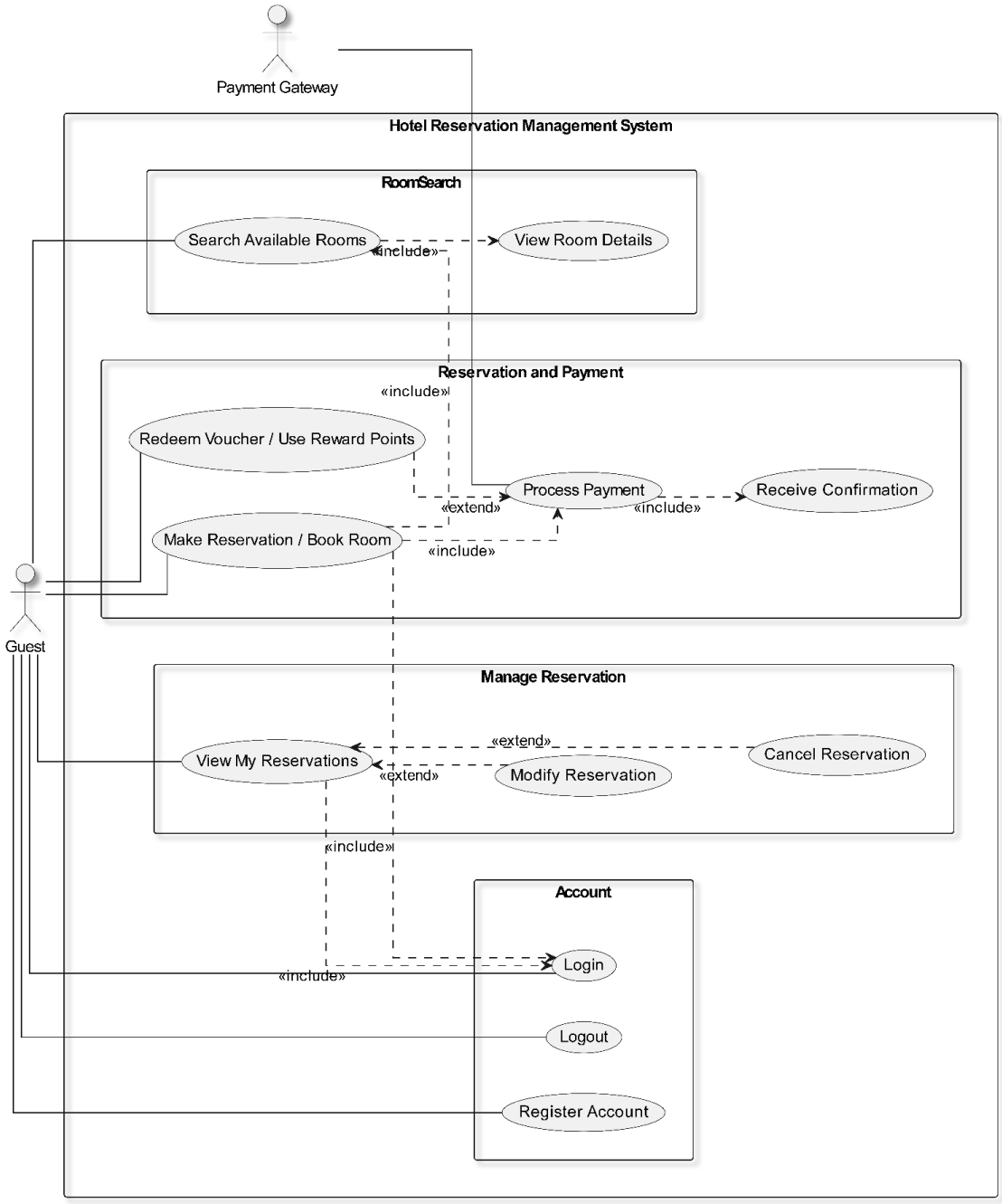

Figure 2: Use Case for Guest and Payment Gateway.

Login	
Use Case Description	The guest logs into the system.
Actor	Guest
Precondition	The Guest has a registered account.
Main Flow	1. The Guest selects Login. 2. The system displays the login form. 3. The Guest enters email and password. 4. The system verifies the credentials. 5. The system grants access and opens the Guest dashboard.
Postcondition	The Guest is authenticated and an active session is created.

Table 1: Login.

Register Account	
Use Case Description	Guest creates a new account in the system.
Actor	Guest
Precondition	The Guest is not logged in and does not already have an account.
Main Flow	1. The Guest selects Register Account. 2. The system displays the registration form. 3. The Guest enters the required information (e.g., name, email, password). 4. The system validates the data. 5. The system creates a new account and stores it in the database. 6. The system confirms successful registration.
Postcondition	A valid Guest account is created and saved.

Table 2: Register Account.

Search Available Rooms	
Use Case Description	Guest searches for rooms based on preferred dates and criteria.
Actor	Guest
Precondition	The system is online.
Main Flow	1. The Guest enters search criteria (check-in/check-out dates, number of guests, room type, etc.). 2. The system validates the entered criteria. 3. The system searches the database for matching available rooms. 4. The system displays a list of available rooms.
Postcondition	A list of available rooms (or a “no results” message) is displayed.

Table 3: Search.

View Room Details	
Use Case Description	Guest views detailed information about a selected room.
Actor	Guest
Precondition	Search results are available.
Main Flow	1. The Guest clicks on a specific room listing. 2. The system retrieves detailed information about the room. 3. The system displays room details (price, features, capacity, images, policies).
Postcondition	Room details are visible to the Guest.

Table 4: View.

Make Reservation / Book Room	
Use Case Description	Guest books a selected room.
Actor	Guest
Precondition	1. The Guest is logged in. 2. A room has been selected and is still available.
Main Flow	1. The Guest selects Make Reservation / Book Room. 2. The system shows the reservation summary (room, dates, price). 3. The Guest confirms the reservation details. 4. The system rechecks room availability. 5. The system creates a reservation with status Pending Payment. 6. The system directs the Guest to payment.
Postcondition	A reservation is created and awaits payment.

Table 5: Reservation.

Process Payment	
Use Case Description	Guest pays for the reservation through the Payment Gateway.
Actor	Guest
Precondition	A reservation exists with status Pending Payment.
Main Flow	1. The Guest selects a payment method. 2. The system forwards payment data to the Payment Gateway. 3. The Guest enters payment details. 4. The Payment Gateway processes the payment. 5. The Payment Gateway returns a successful response. 6. The system updates the reservation status to Paid/Confirmed.
Postcondition	Payment is completed and the reservation becomes confirmed.

Table 6: Process Payment.

Receive Confirmation	
Use Case Description	System sends booking confirmation to Guest.
Actor	Guest
Precondition	Payment is successful.
Main Flow	1. The system generates confirmation (reservation ID, details, dates, room info). 2. The system sends confirmation via app/email/SMS. 3. The Guest receives and views the confirmation.
Postcondition	The Guest has a confirmed reservation and proof of booking.

Table 7: Receive Confirmation.

Redeem Voucher / Use Reward Points	
Use Case Description	Guest applies voucher or reward points during payment.
Actor	Guest
Precondition	1. The Guest is logged in. 2. A reservation is in progress and payment has not yet been finalized. 3. The Guest has a valid voucher or sufficient reward points.
Main Flow	1. The Guest chooses to redeem a voucher or points. 2. The system verifies validity and available balance. 3. The system applies the discount to the total amount. 4. The system displays the updated payable amount.
Postcondition	The amount is adjusted before payment.

Table 8: Use Reward Points.

View My Reservations	
Use Case Description	Guest views all current and past reservations.
Actor	Guest
Precondition	The Guest is logged in.
Main Flow	1. The Guest selects View My Reservations. 2. The system retrieves the Guest's reservations from the database. 3. The system displays current and past reservations.
Postcondition	The Guest can view their reservation history.

Table 9: View My Reservations.

Modify Reservation	
Use Case Description	Guest updates an existing reservation.
Actor	Guest
Precondition	1. The Guest is logged in. 2. The Guest has an active reservation eligible for modification.
Main Flow	The Guest opens View My Reservations. The Guest selects a reservation and clicks Modify Reservation. The system displays modifiable fields (dates, room type, etc.). The Guest submits new details. The system validates availability and policy rules. The system updates the reservation and shows success.
Postcondition	The reservation is updated with new details.

Table 10: Modify Reservation.

Cancel Reservation	
Use Case Description	Guest cancels an existing reservation.
Actor	Guest
Precondition	1. The Guest is logged in. 2. The Guest has an active reservation eligible for cancellation.
Main Flow	1. The Guest opens View My Reservations. 2. The Guest selects a reservation and clicks Cancel Reservation. 3. The system displays cancellation policy and confirmation prompt. 4. The Guest confirms cancellation. 5. The system cancels the reservation and updates room availability. 6. The system shows cancellation success.
Postcondition	The reservation is cancelled and availability is updated.

Table 11: Cancel Reservation.

Logout	
Use Case Description	Guest logs out from the system.
Actor	Guest
Precondition	The Guest is logged in.
Main Flow	1. The Guest chooses Logout. 2. The system terminates the active session. 3. The system redirects the Guest to the home page.
Postcondition	The Guest is logged out successfully.

Table 12: Logout.

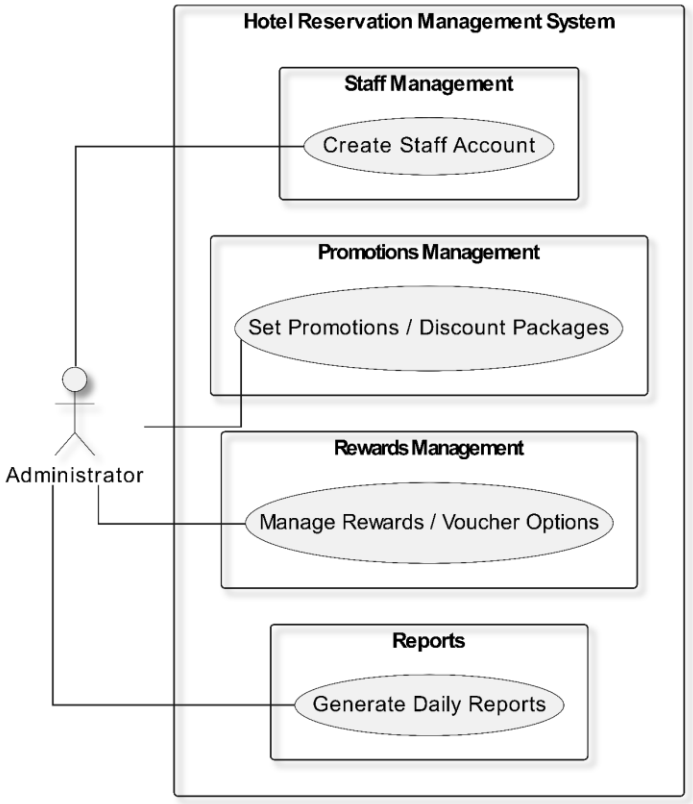

Figure 3: Use Case for Administrator.

Create Staff Account	
Use Case Description	Administrator creates a new staff account in the system.
Actor	Administrator
Precondition	The Administrator is logged in with valid admin privileges
Main Flow	1. The Administrator opens the staff management section. 2. The Administrator selects Create Staff Account. 3. The system displays a staff registration form. 4. The Administrator enters staff details (name, role, contact, credentials). 5. The system validates the information entered. 6. The system creates the staff account and stores it. 7. The system confirms successful account creation.
Postcondition	A new staff account is created and ready for use.

Table 13: Create Staff Account.

Set Promotions / Discount Packages	
Use Case Description	Administrator creates or updates promotions and discount packages.
Actor	Administrator
Precondition	The Administrator is logged in.
Main Flow	1. The Administrator opens the promotions management section. 2. The Administrator selects Set Promotions / Discount Packages. 3. The system displays promotion configuration options. 4. The Administrator specifies promotion rules (type, discount value, dates, conditions). 5. The system validates the promotion details. 6. The system activates and stores the promotion. 7. The system confirms the promotion is applied.
Postcondition	A new promotion/discount package is active in the system.

Table 14: Set Promotions.

Manage Rewards / Voucher Options	
Use Case Description	Administrator manages reward points rules and voucher options.
Actor	Administrator
Precondition	The Administrator is logged in.
Main Flow	1. The Administrator opens the rewards management section. 2. The system shows current reward rules and voucher options. 3. The Administrator adds, edits, or removes reward/voucher configurations. 4. The system validates the updates. 5. The system saves the new reward/voucher settings. 6. The system confirms successful updates.
Postcondition	Rewards and voucher options are updated and applied system wide.

Table 15: Manage Rewards.

Generate Daily Reports	
Use Case Description	Administrator generates daily operational reports.
Actor	Administrator
Precondition	The Administrator is logged in.
Main Flow	1. The Administrator selects Generate Daily Reports. 2. The system gathers daily operational data (reservations, payments, occupancy, etc.). 3. The system generates the daily report. 4. The system displays the report and provides an export/download option.
Postcondition	A daily report is generated and available for review/export.

Table 16: Generate Daily Reports.

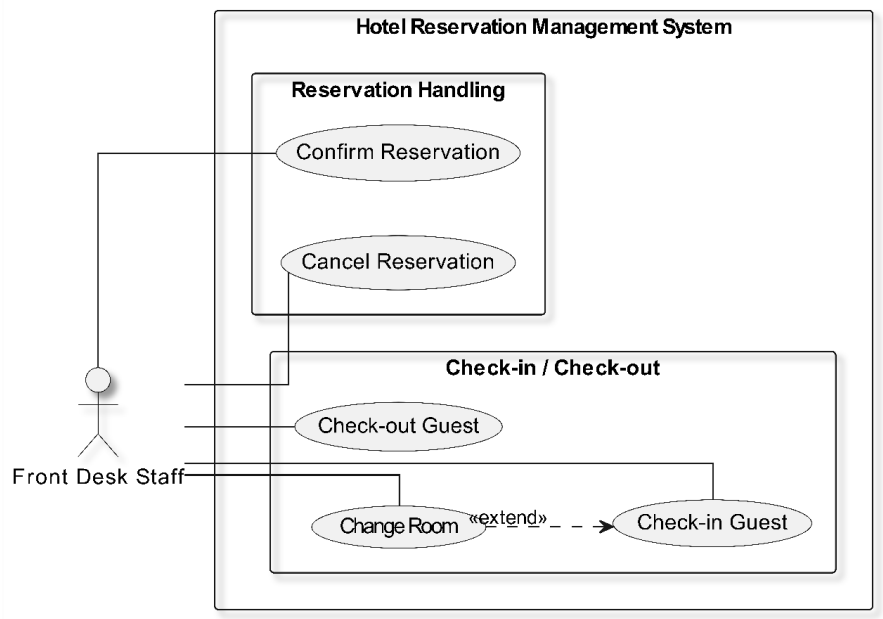

Figure 4: Use Case for Front Desk Staff.

Confirm Reservation	
Use Case Description	Front Desk Staff confirms a guest reservation.
Actor	Front Desk Staff
Precondition	The staff member is logged in; a reservation exists.
Main Flow	The staff member searches for the reservation. The system displays reservation details. The staff member selects Confirm Reservation. The system updates the reservation status to Confirmed. The system shows confirmation success.
Postcondition	Reservation is confirmed successfully.

Table 17: Confirm Reservation.

Cancel Reservation	
Use Case Description	Front Desk Staff cancels a guest reservation.
Actor	Front Desk Staff
Precondition	Staff is logged in; the reservation exists and is cancellable
Main Flow	The staff member opens the reservation record. The staff member selects Cancel Reservation. The system displays the cancellation policy. The staff member confirms cancellation. The system cancels the reservation and updates availability. The system shows cancellation success.
Postcondition	Reservation is cancelled, and room availability is updated.

Table 18: Cancel Reservation.

Check-in Guest	
Use Case Description	Front Desk Staff checks in a guest with a valid reservation.
Actor	Front Desk Staff
Precondition	Staff is logged in. Guest has confirmed reservation.
Main Flow	The staff locates the guest reservation. Staff verifies guest identity. Staff selects Check-in Guest. The system updates the reservation status to Checked-in. The system confirms check-in.
Postcondition	Guest is checked in.

Table 19: Check-in Guest.

Check-out Guest	
Use Case Description	Front Desk Staff checks out a guest.
Actor	Front Desk Staff
Precondition	Guest is currently checked in
Main Flow	Staff opens the guest's reservation. Staff confirms checkout request. The system verifies all charges are settled. Staff selects Check-out Guest. The system updates status to Checked-out. The system releases the room.
Postcondition	Guest is checked out and room becomes available/dirty for cleaning.

Table 20: Check-out Guest.

Change Room	
Use Case Description	Front Desk Staff changes the guest's assigned room.
Actor	Front Desk Staff
Precondition	Guest is checked in; another room is available
Main Flow	Staff opens the guest reservation. Staff selects Change Room. The system displays available alternative rooms. Staff selects a new room. The system updates the reservation and room allocations. The system confirms the room change.
Postcondition	Guest is reassigned to a new room and records are updated

Table 21: Change Room.

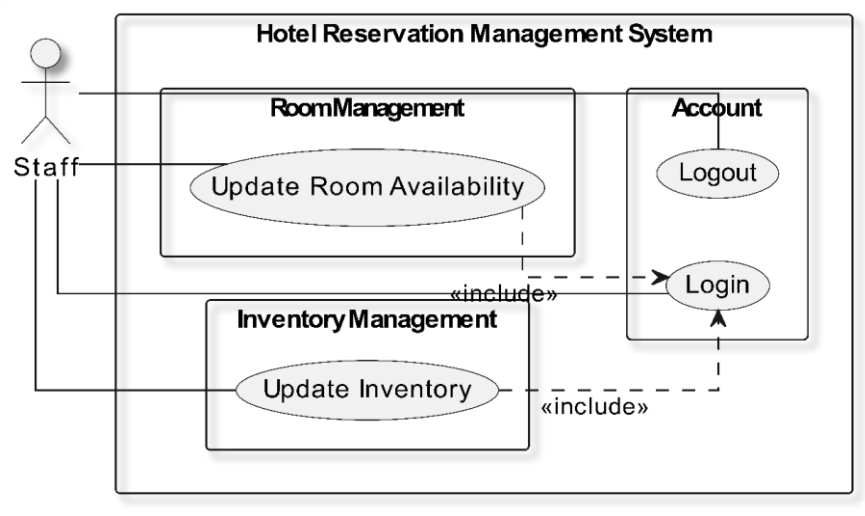

Figure 5: Use Case for Staff.

Update Room Availability	
Use Case Description	Staff updates room availability status in the system.
Actor	Staff
Precondition	Staff is logged in
Main Flow	Staff selects Update Room Availability. The system displays rooms and current availability. Staff updates a room's availability status. The system validates and saves the update. The system confirms the change.
Postcondition	Room availability is updated in the system

Table 22: Update Room Availability.

Update Inventory	
Use Case Description	Staff updates hotel inventory levels.
Actor	Staff
Precondition	Staff is logged in
Main Flow	Staff selects Update Inventory. The system displays inventory items and quantities. Staff modifies quantities or item details. The system validates the inputs. The system saves updates and confirms success.
Postcondition	Inventory data is updated

Table 23: Update Inventory.

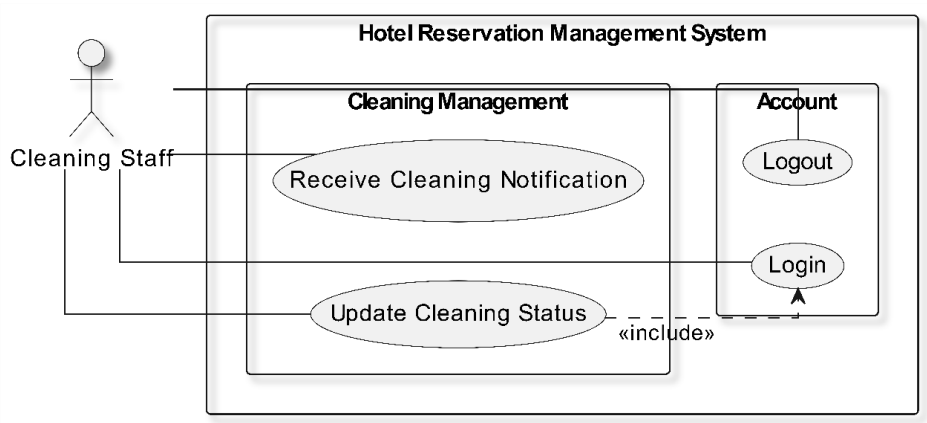

Figure 6: Use Case for Cleaning Staff.

Receive Cleaning Notification	
Use Case Description	Cleaning Staff receives a notification to clean a room.
Actor	Cleaning Staff
Precondition	Cleaning staff is logged in.
Main Flow	The system detects a room needing cleaning (e.g., after checkout). The system sends a cleaning notification to Cleaning Staff. Cleaning Staff receives and views the notification.
Postcondition	Cleaning task is assigned/visible to staff.

Table 24: Receive Cleaning Notification.

Update Cleaning Status	
Use Case Description	The cleaning staff updates the cleaning progress/status of a room.
Actor	Cleaning Staff
Precondition	Cleaning staff is logged in and has an assigned task
Main Flow	Cleaning Staff opens the assigned cleaning task. Cleaning Staff updates the room status (e.g., In Progress / Cleaned). The system validates the update. The system saves the new cleaning status. The system confirms the update.
Postcondition	Room cleaning status is updated in the system.

Table 25: Update Cleaning Status.

4.2 Sequence Diagrams with Descriptions

1. Guest: Online Room Reservation

This sequence diagram depicts the interaction scenario between a guest and the HRMS system for making online room reservations. The process initiates with guest authentication, then it proceeds to room searching based on date range selection and number of guests. After room selection, it moves into the payment stage, where reward points or vouchers can optionally be applied. The system processes the payment through a secure external payment-processing gateway. If the payment is successful, the reservation is saved in the database, the rooms get updated for availability, and confirmation to the guest is sent via email or SMS. Hence, the system reflects updated information about the reservation on the guest dashboard.

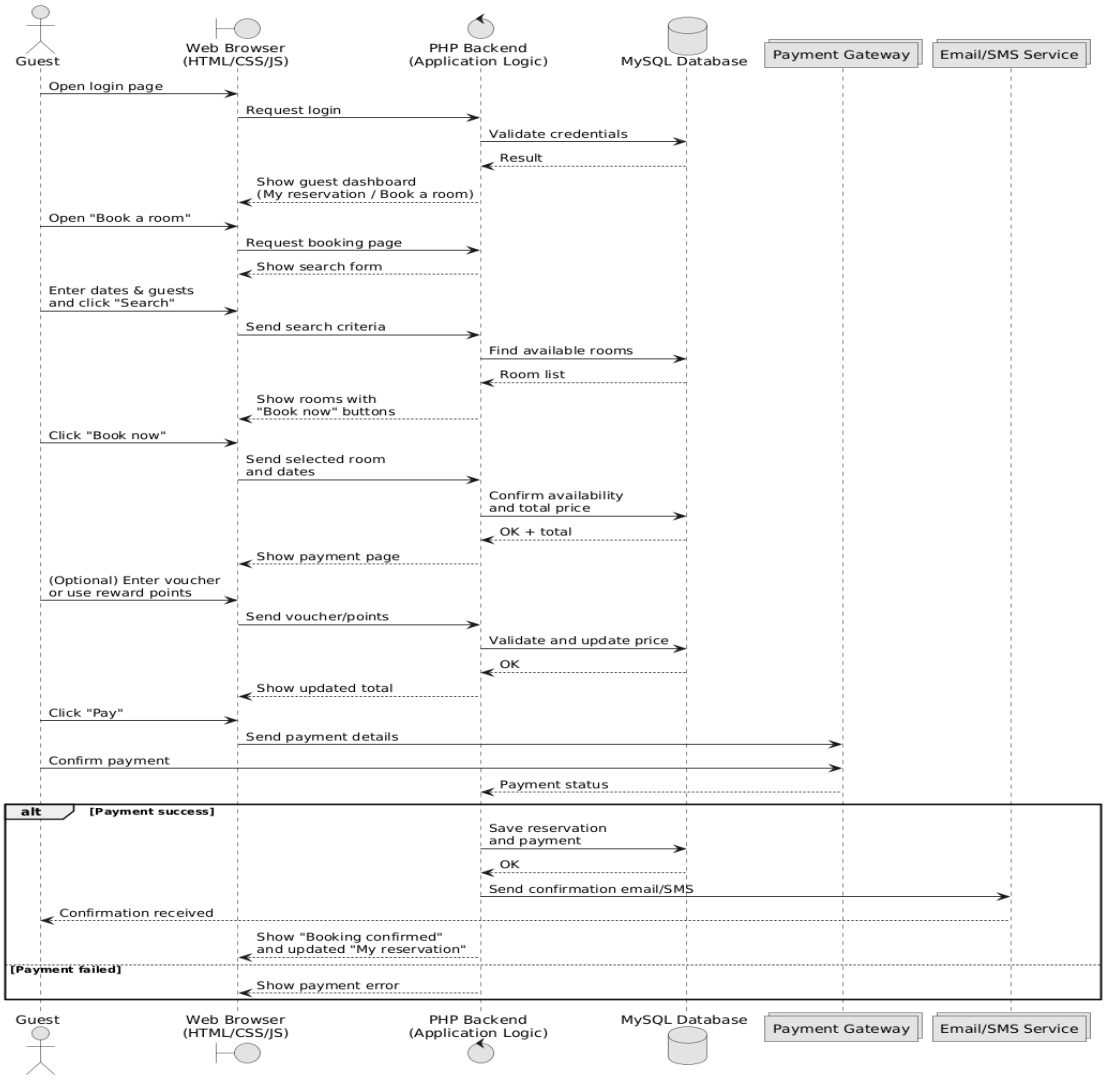

Figure 7: Guest sequence diagram.

2. Administrator: Manage Promotions & Staff

This sequence diagram describes the administrator's interaction with the HRMS over promotional offers and staff accounts. On successful login into the system, an administrator can enter the promotions module to list all existing ones, create new promotional offers, or deactivate outdated ones. The system saves every update in the database and refreshes the promotions list at once. Moreover, the administrator is able to manage the staff accounts by entering the module of staff management, where all registered staff can be viewed and

new staff can be added by submitting information about them. The backend picks this up, validates, and saves this new staff information into the database. All changes are instantly reflected in the UI, ensuring appropriate logging and visibility of the actions taken by an administrator across the system.

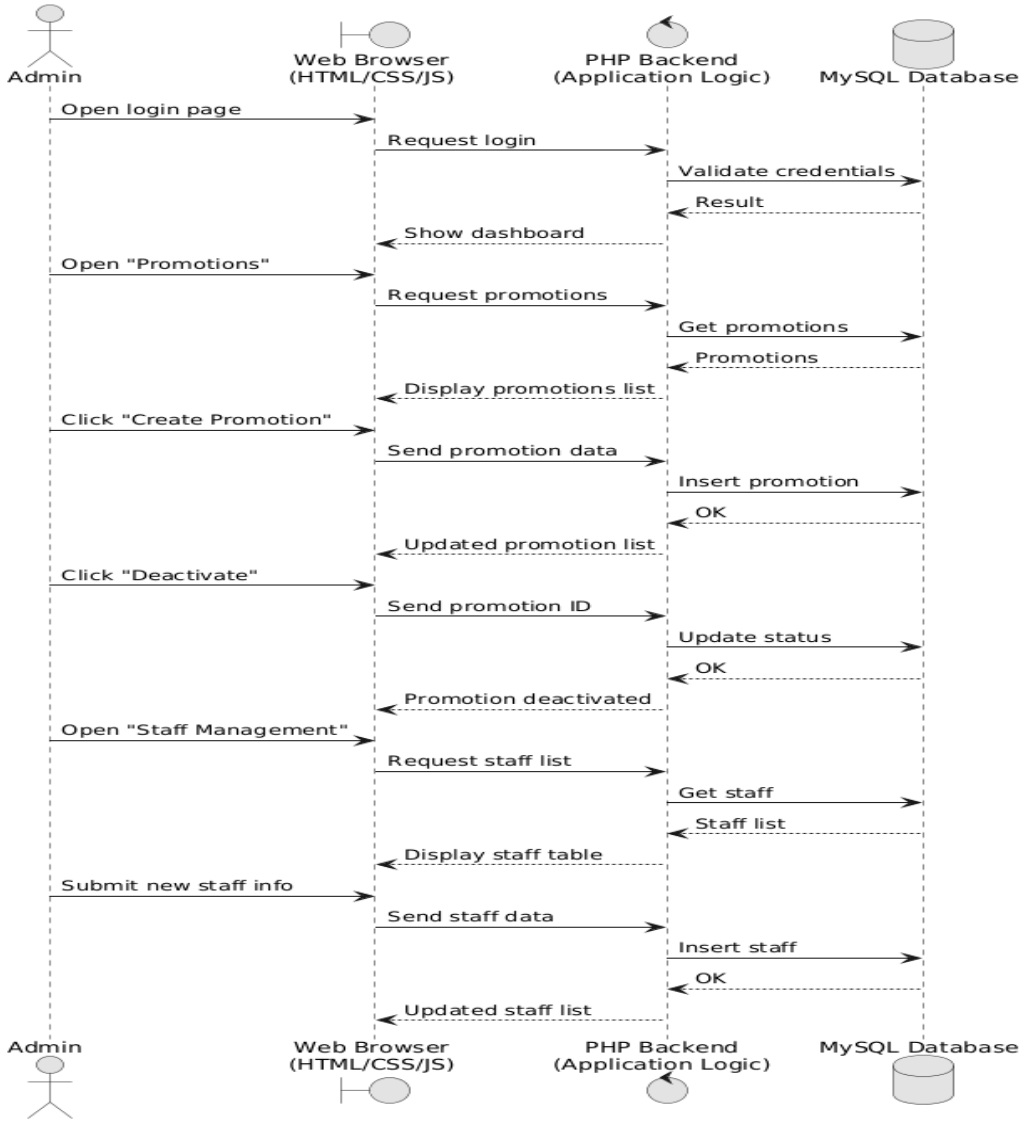

Figure 8: Admin sequence diagram.

3. Front Desk Staff*:* Check-in / Check-out & Room Change

This sequence diagram presents the management of guest reservations by front desk staff within the HRMS. After logging in, the staff member shall be able to see all currently active reservations. The staff shall perform some key actions, such as checking a guest in, checking a guest out, or updating a reservation by assigning a new room to them. Each of these actions will fire appropriate updates in the database to accurately reflect the reservation and room status. Then, the interface displays the updated information for the staff member to confirm.

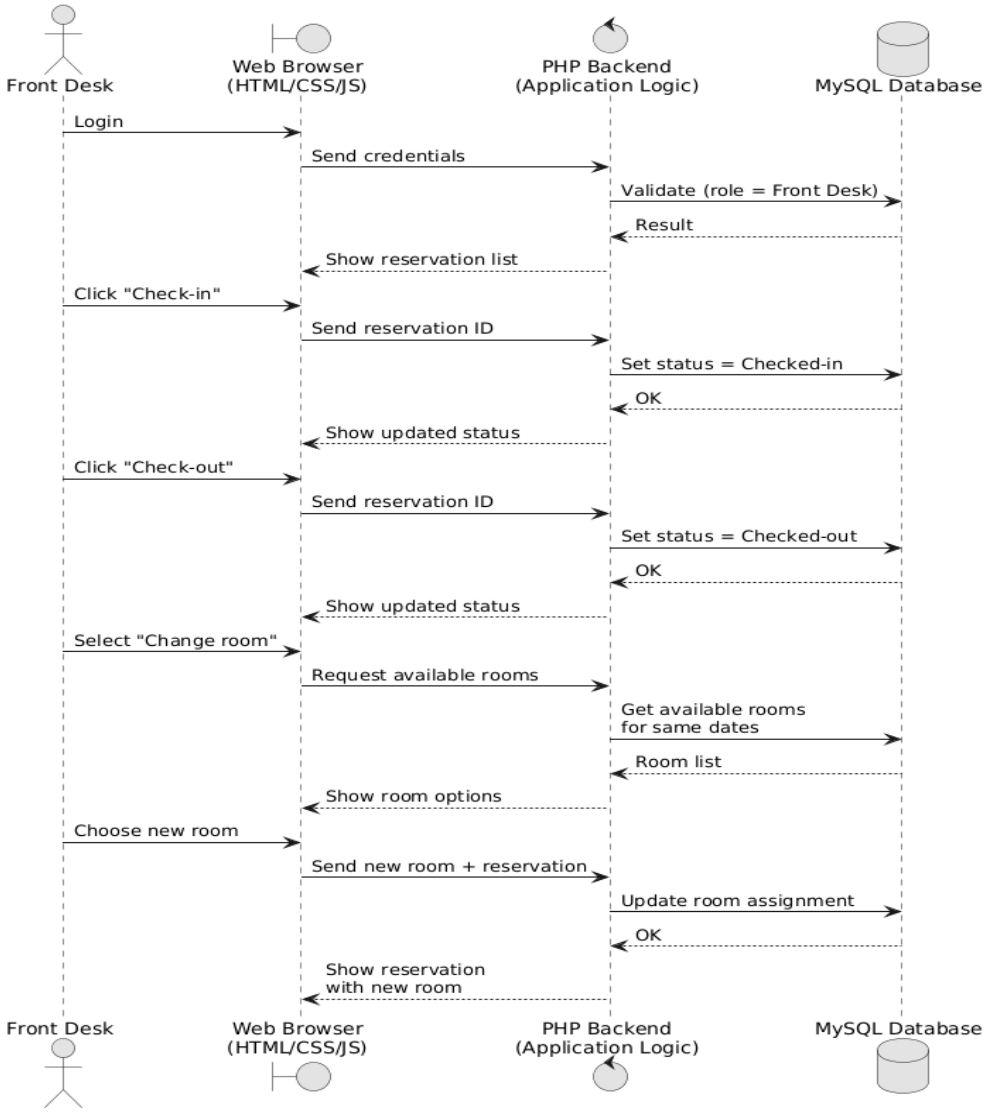

Figure 9: Front Desk Staff sequence diagram.

4. Staff: Update Inventory & Room Availability

This sequence diagram represents the workflow for staff members responsible for inventory and room status updates. Once authenticated, staff can view inventory items and adjust their quantities, such as linens, towels, or toiletries. They will also be able to view room availability and update the statuses of rooms: Available, Occupied, or Maintenance. Every modification is then saved in the database and instantly reflected in the user interface to assure valid operational data throughout the hotel.

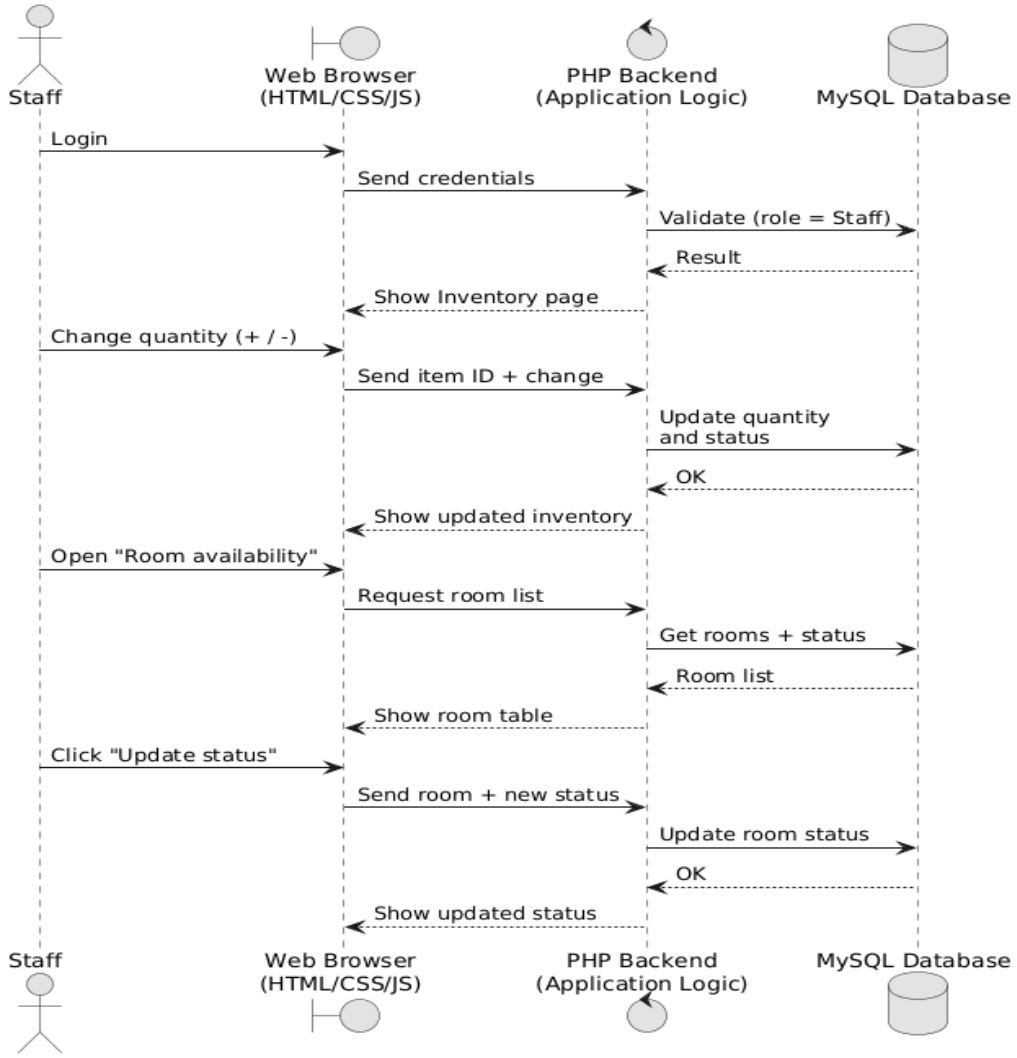

Figure 10: Staff sequence diagram.

5. Cleaning Staff: Housekeeping Workflow

The given flowchart depicts the housekeeping work process in relation to cleaning personnel. Following system login, the cleaner is presented with a list containing the housekeeping tasks assigned. A cleaner can initiate the task by marking the room status "In Progress" and complete the task by changing the room status to "Completed." All status updates are saved into the database and instantly updated in the user interface to provide the actual status of the cleanliness of rooms in real time for the staff and administrators.

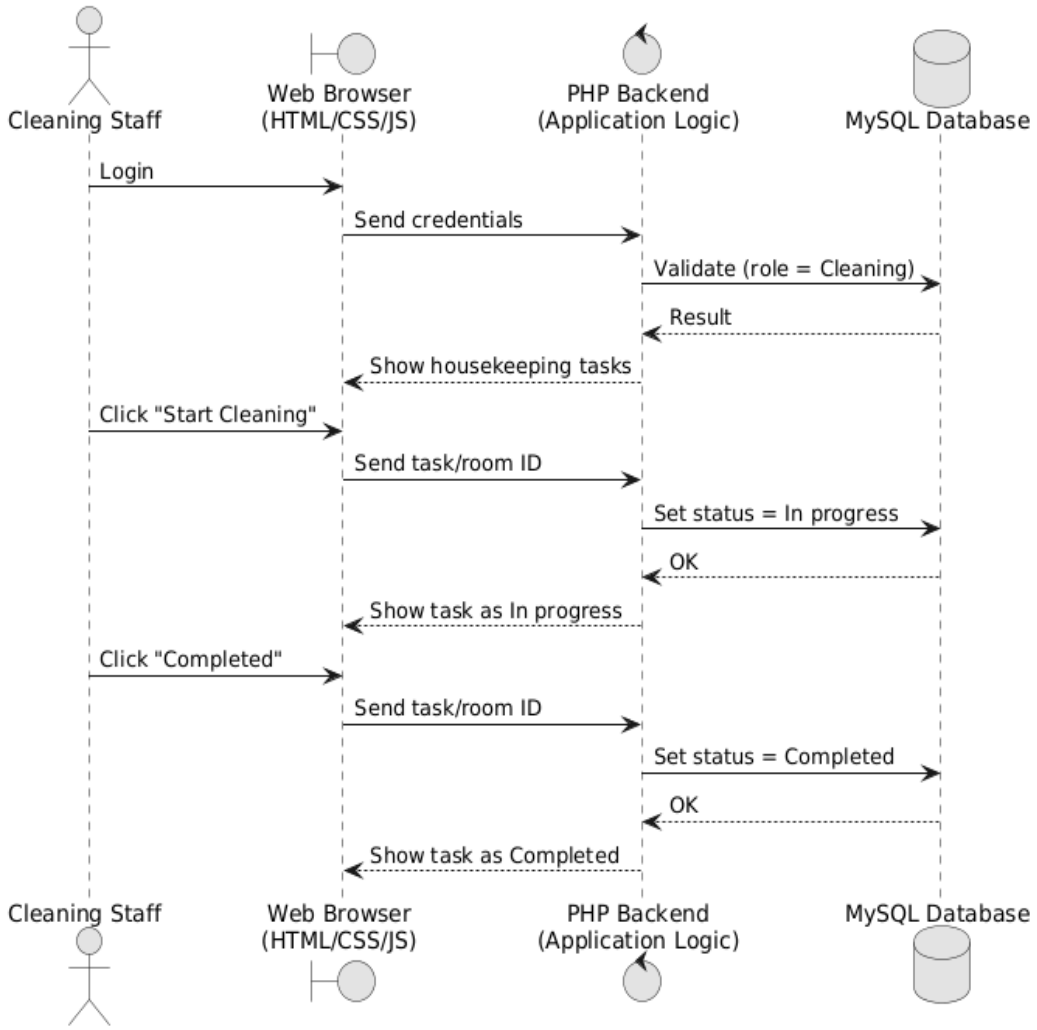

Figure 11: Cleaning staff sequence diagram.

4.3 Activity Diagram with Description

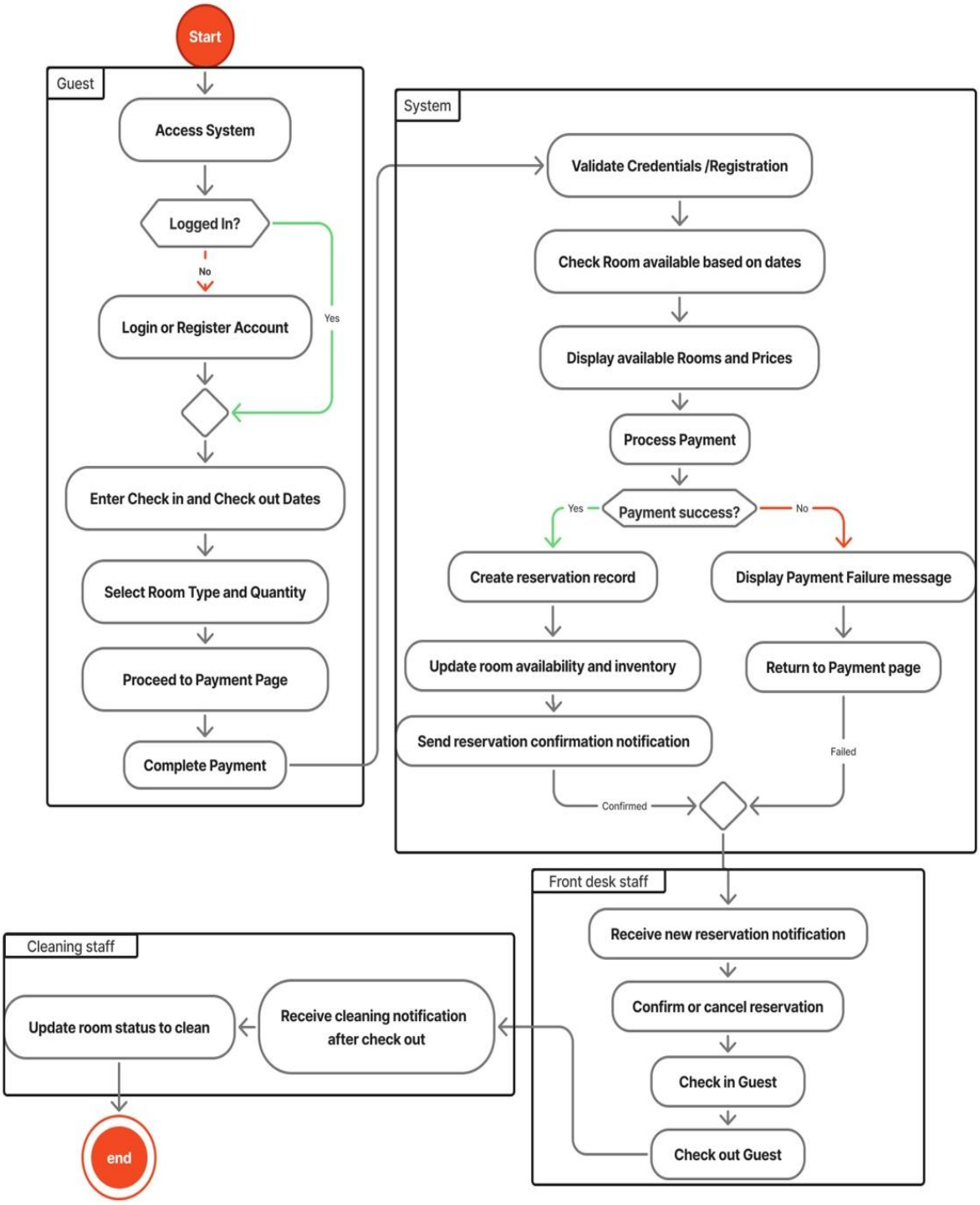

Figure 12: Activity Diagram.

Completing a hotel room reservation	
Activity Description	Completing a hotel room reservation through the system.
Actor	Guest
Precondition	1. The system is running and accessible. 2. The Guest has the intent to reserve a room
Main Flow	1. Guest accesses the system 2. System check if the Guest is logged in 3. Guest if not logged in so should logs in or registers a new account 4. Guest enters the desired check in and check out dates 5. Guest specifies the room type and quantity 6. System checks room availability based on the entered criteria 7. System displays the available rooms and price 8. Guest selects a suitable room and proceeds to the payment page 9. Guest completes the payment process 10. System processes and validates the payment 11. System if payment is successful so creates the reservation record, updates room availability, and sends a confirmation notification to the Guest
Postcondition	The booking record is created, room availability is updated, and then the user receives a confirmation message.

Table 26: Activity Description.

4.4 Context Diagram with Description

The context diagram illustrates the overall interaction between the Hotel Reservation Management System and external entities involved in hotel operations. The system serves as the central hub that manages room reservations, payments, staff activities, and administrative functions. Guests interact with the system to register, log in, search rooms, make bookings, and complete payments, while receiving confirmation, room availability updates, and payment status notifications. Front Desk Staff use the system to manage guest reservations, perform check-in and check-out operations, and access reservation details. The Administrator oversees system management by handling promotions, staff accounts, and generating reports. Staff members manage room inventory and update room status, whereas Cleaning Staff update room cleaning status and receive assigned room information. The system also communicates with the Bank to process payments through validation and transaction requests. Overall, the diagram provides a high-level view of data flow between the system and its key actors, demonstrating how information is exchanged to support hotel reservation processes.

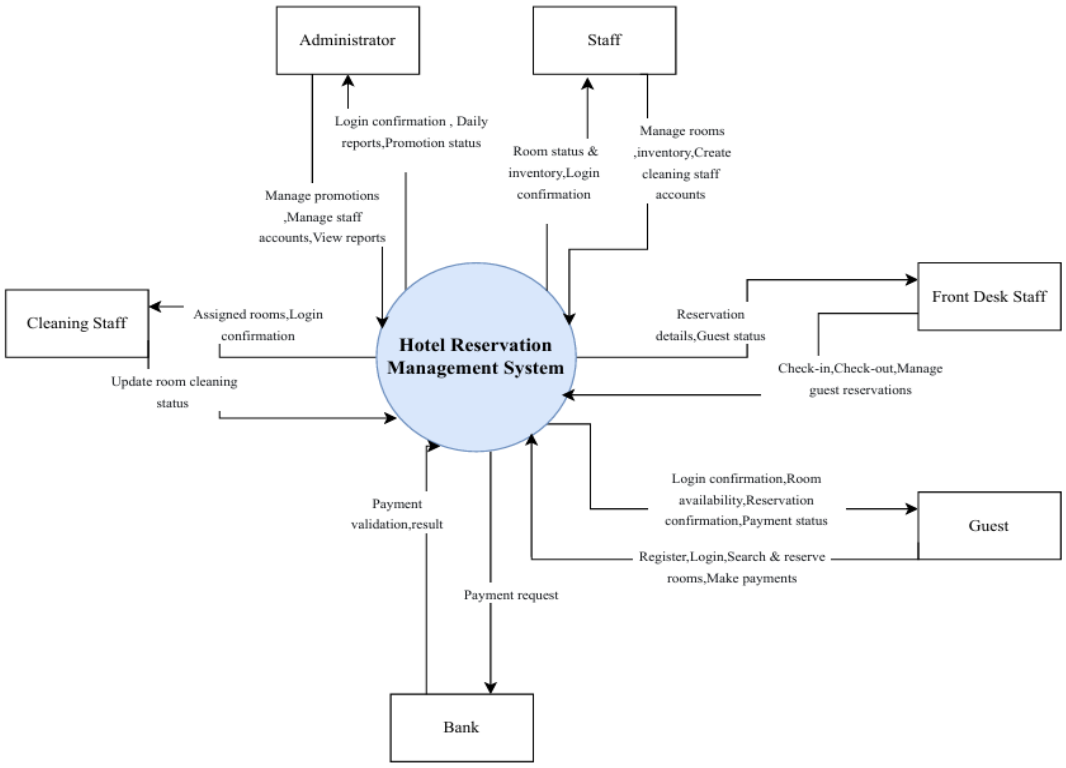

Figure 13: Context Diagram.

6. Prototype

The prototype is an early working model of the Hotel Reservation System, showcasing core functions such as login, room search, availability display, and booking. It allows users to experience the basic workflow, visualize system operation, and provide feedback before full implementation.

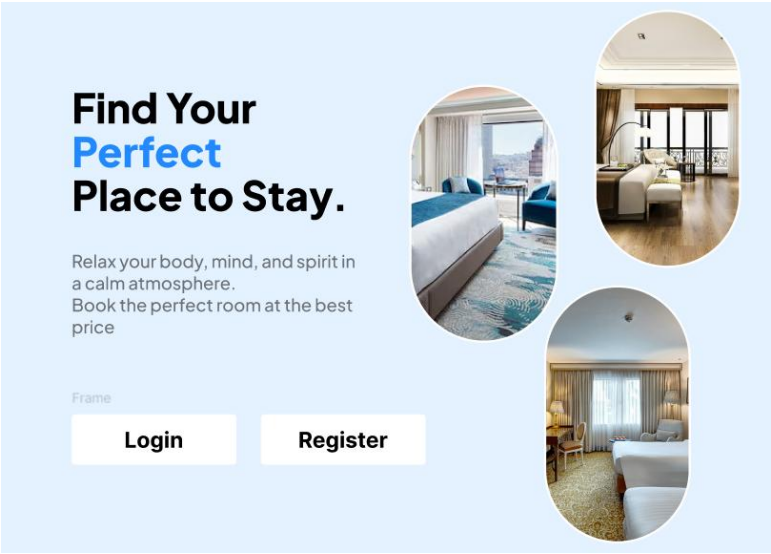

Figure 14: Home Page.

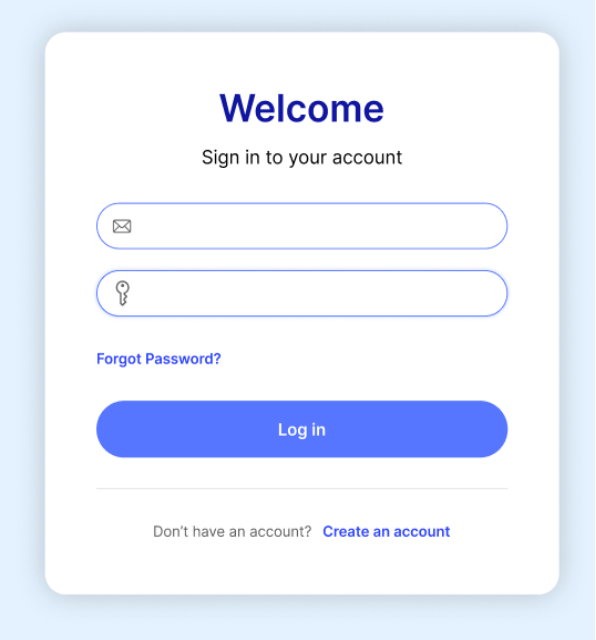

Figure 15: Login Page.

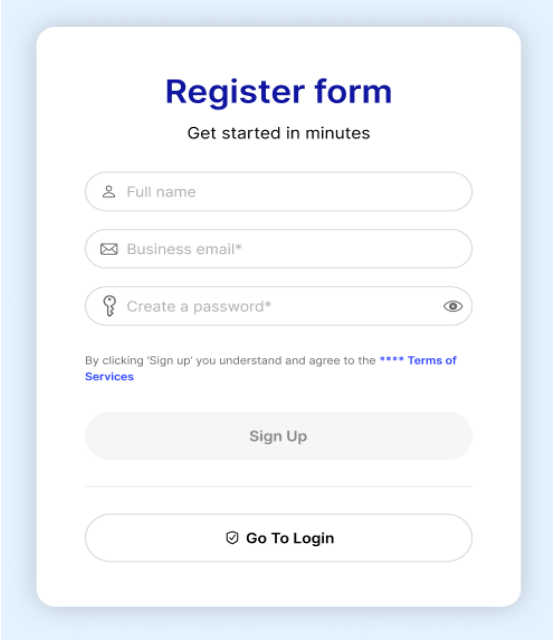

Figure 16: Registration Page.

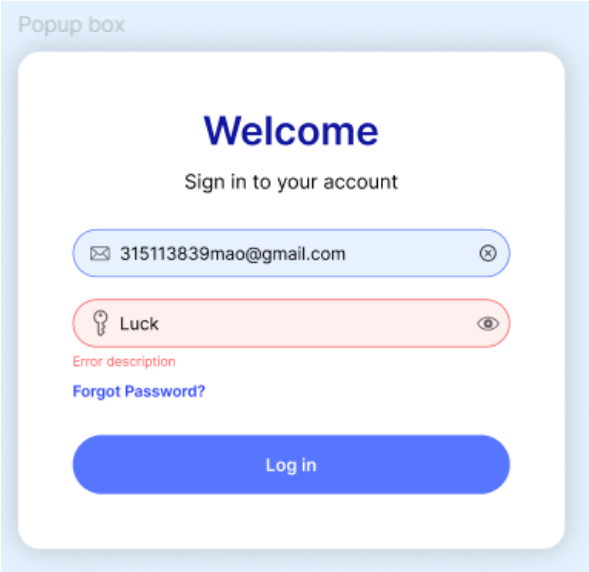

Figure 17: Displaying an invalid password error during login.

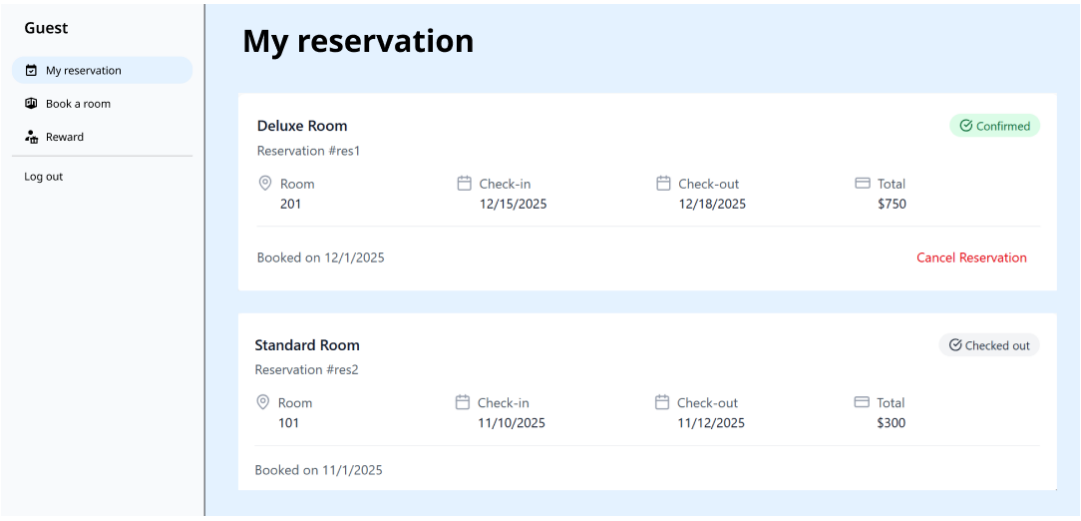

Figure 18: Reservation page showing current and past bookings.

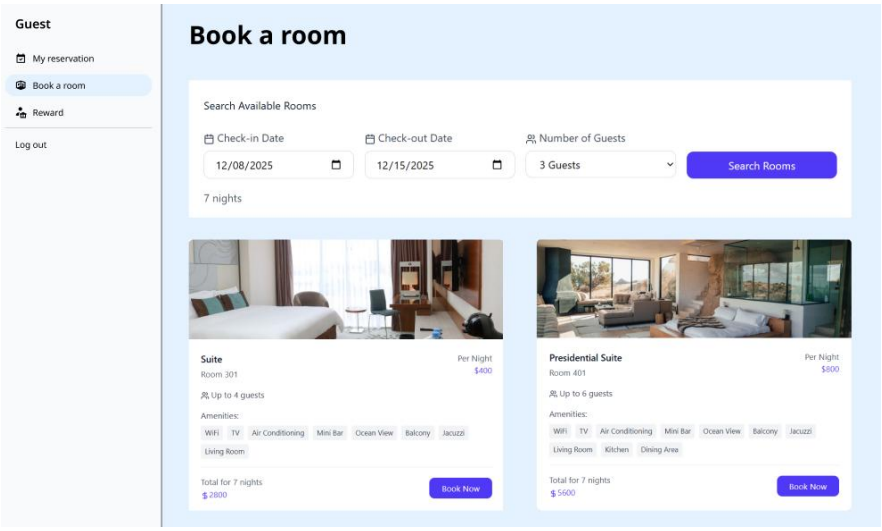

Figure 19: Room booking page showing available rooms, dates, and prices.

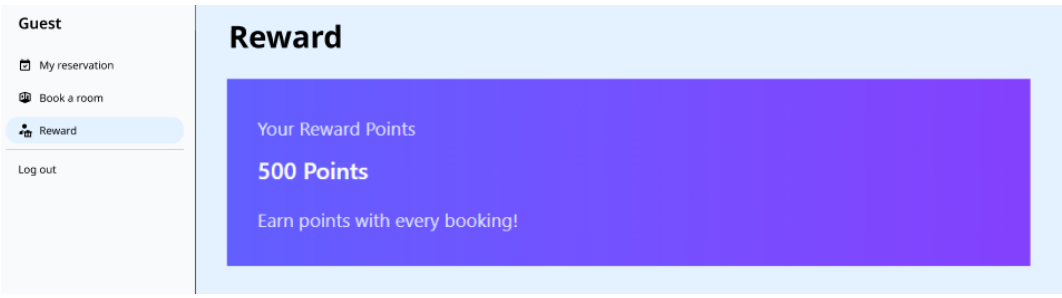

Figure 20: Reward page showing accumulated points.

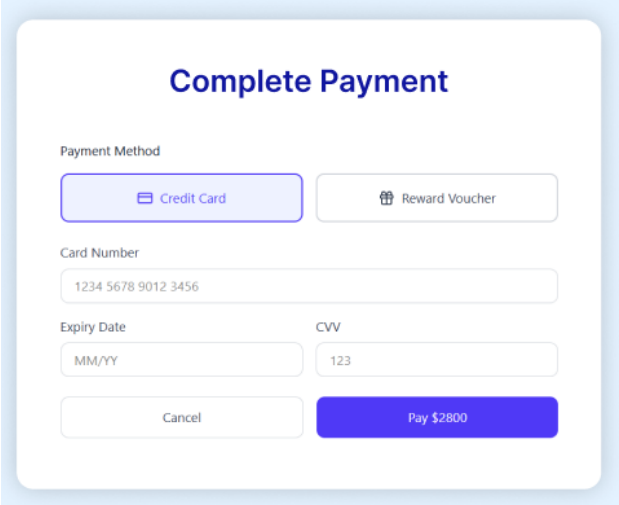

Figure 21: Payment screen for completing a room reservation.

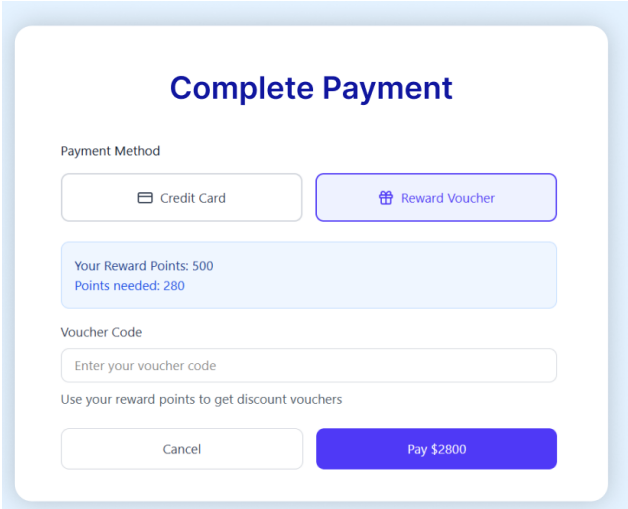

Figure 22: Payment page using reward voucher.

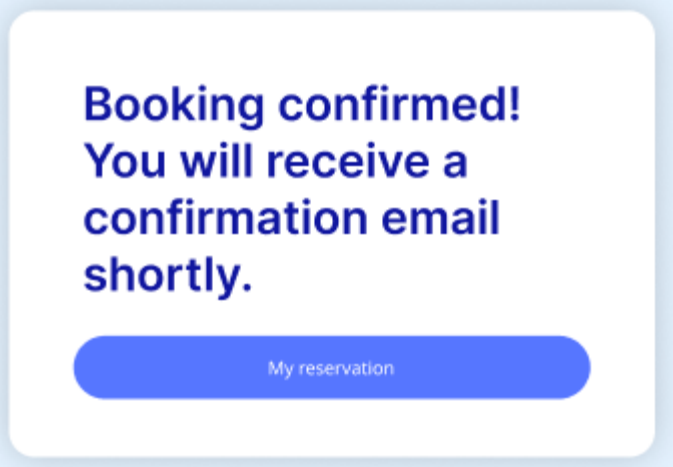

Figure 23: Booking confirmation message.

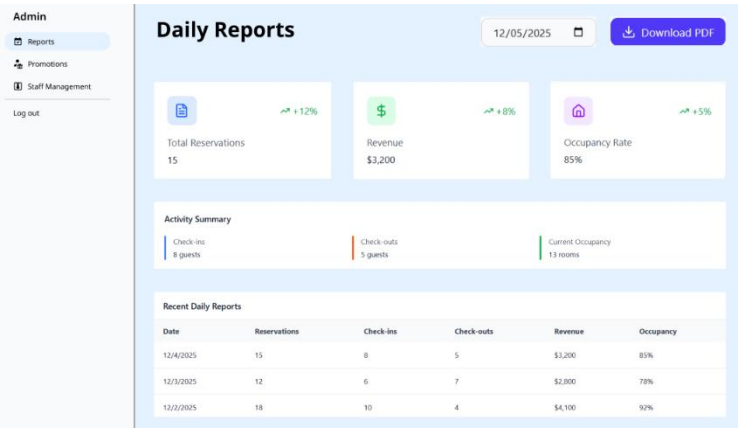

Figure 24: Admin dashboard displaying daily hotel performance reports.

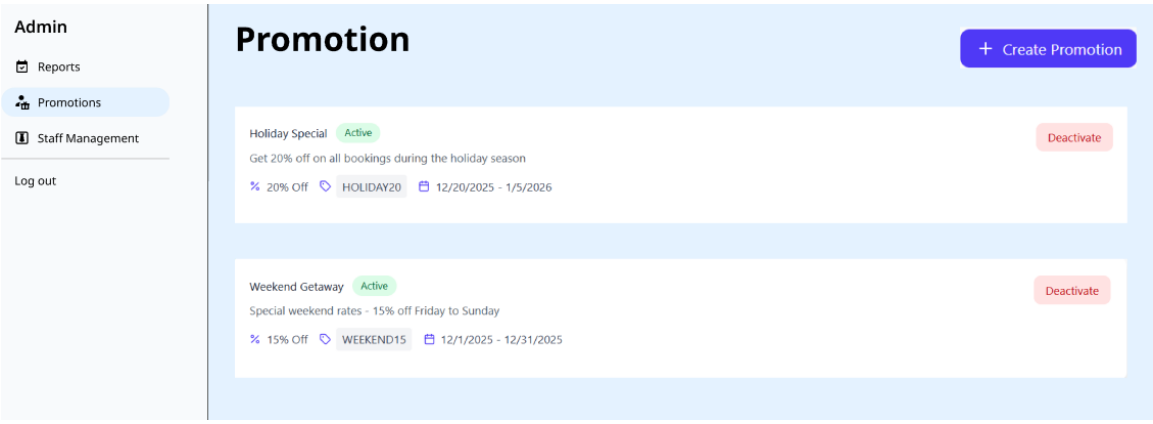

Figure 25: Admin page for managing active promotions.

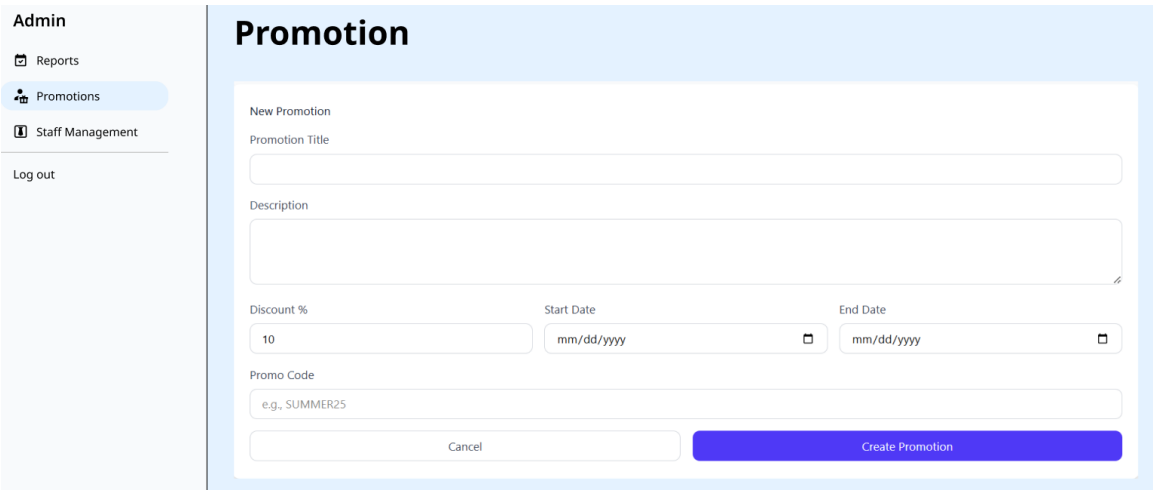

Figure 26: Admin interface for creating new promotions.

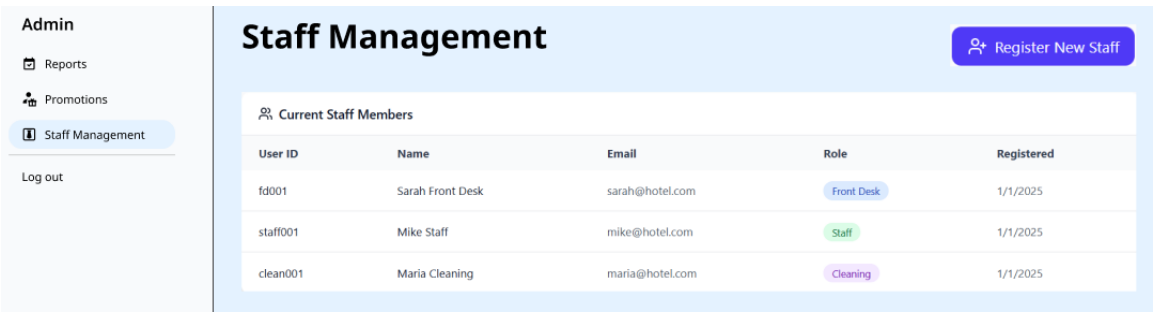

Figure 27: Admin page for managing hotel staff.

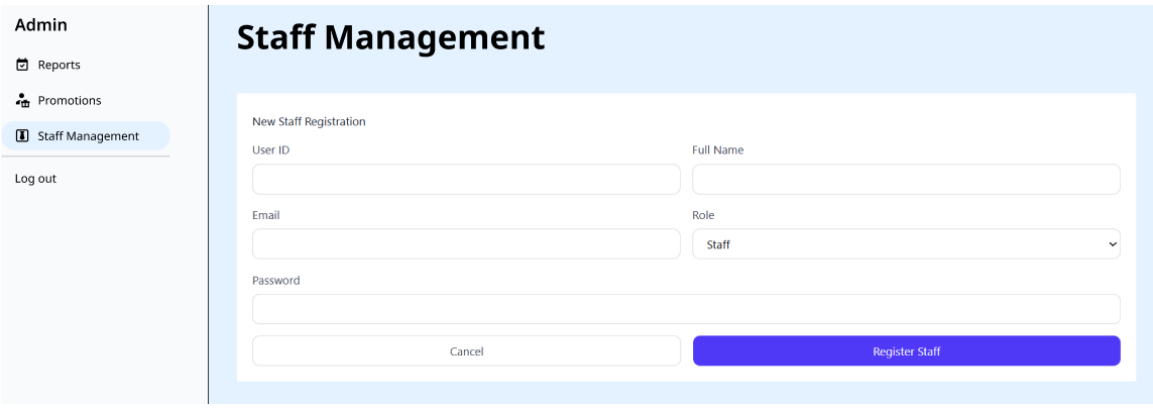

Figure 28: Admin form for registering new staff.

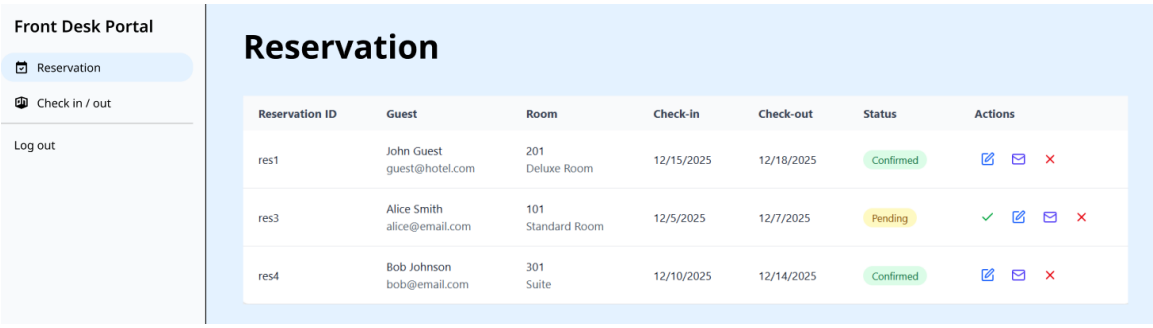

Figure 29: Front desk reservation management page.

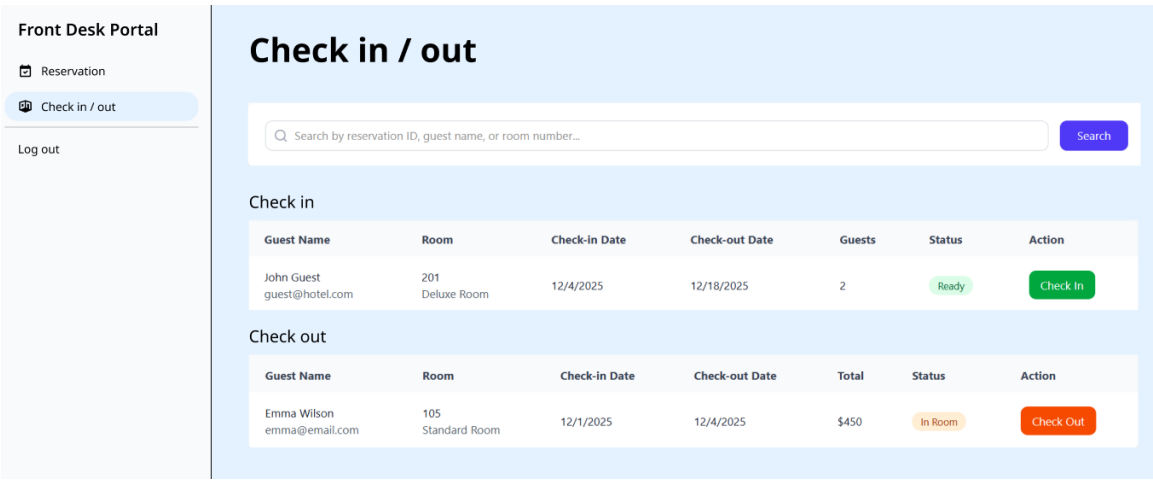

Figure 30: Front desk check-in/check-out management page.

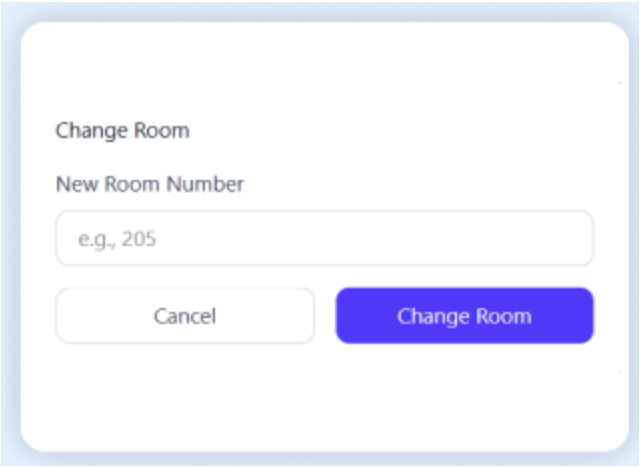

Figure 31: The interface allows staff to update a guest's assigned room number.

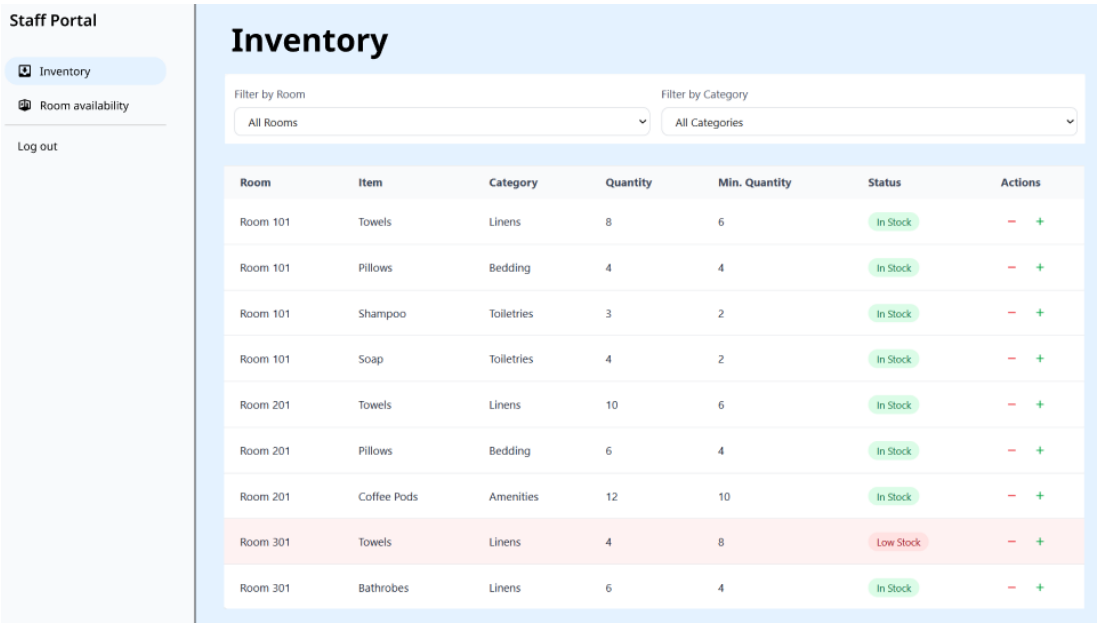

Figure 32: Staff inventory management screen.

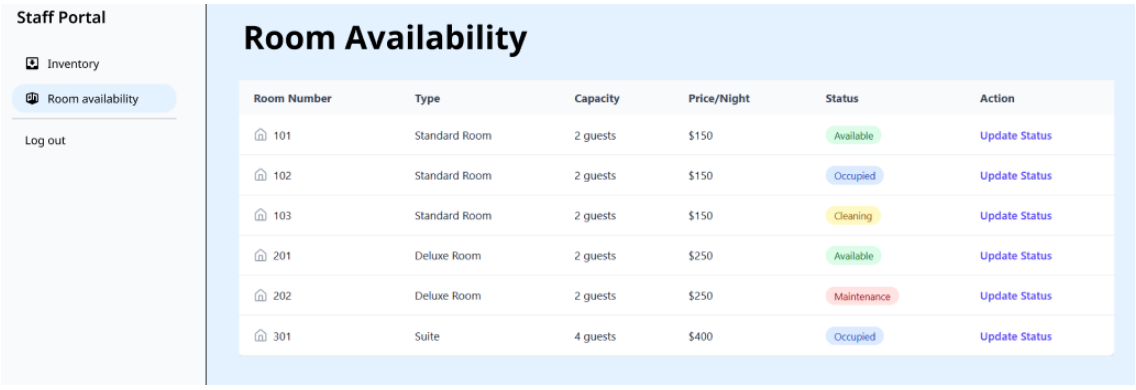

Figure 33: Room Availability interface.

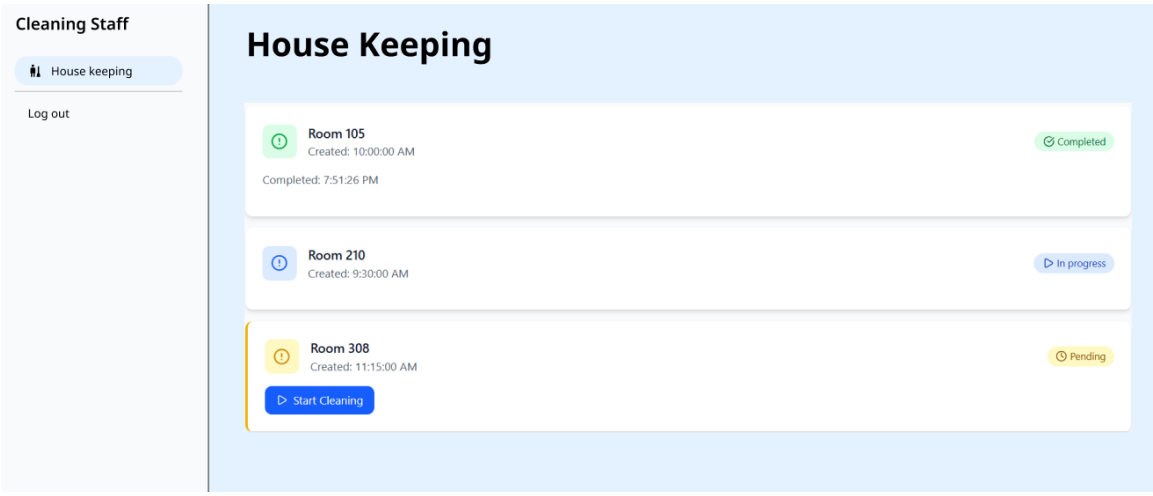

Figure 34: Cleaning staff interface for managing housekeeping tasks.

7. Gantt Chart

A Gantt chart in Figure 35 outlines project tasks, their durations, and dependencies across a timeline to ensure organized project execution.

	Task Mode	Task Name	Duration	Start	Finish	Predecessors
1		Introduction	1 day	Mon 11/10/25	Mon 11/10/25	
2		Methodology Description	7 days	Tue 11/11/25	Wed 11/19/25	
3		Requirement Gathering and Analysis	1 day	Thu 11/20/25	Thu 11/20/25	2
4		System Design and Architecture	2 days	Thu 11/20/25	Fri 11/21/25	2
5		Technology selection and implementation	2 days	Thu 11/20/25	Fri 11/21/25	2
6		Web/Desktop Application Development	2 days	Thu 11/20/25	Fri 11/21/25	2
7		Requirements	4 days	Fri 11/14/25	Wed 11/19/25	
8		User Requirements	1 day	Thu 11/20/25	Thu 11/20/25	7
9		System Requirements	1 day	Thu 11/20/25	Thu 11/20/25	7
10		Functional Requirements	1 day	Thu 11/20/25	Thu 11/20/25	7
11		Non-Functional Requirements	1 day	Thu 11/20/25	Thu 11/20/25	7
12		Architecture Design	3 days	Mon 11/24/25	Wed 11/26/25	4
13		Models	7 days	Tue 11/25/25	Wed 12/3/25	
14		Use Case Diagrams	1 day	Thu 12/4/25	Thu 12/4/25	13
15		Sequence Diagrams	2 days	Thu 12/4/25	Fri 12/5/25	13
16		Activity Diagram	2 days	Thu 12/4/25	Fri 12/5/25	13
17		Context Diagram	2 days	Thu 12/4/25	Fri 12/5/25	13
18		Prototype	3 days	Mon 12/8/25	Wed 12/10/25	
19		Conclusion	1 day	Thu 12/11/25	Thu 12/11/25	
20		References	1 day	Thu 12/11/25	Thu 12/11/25	

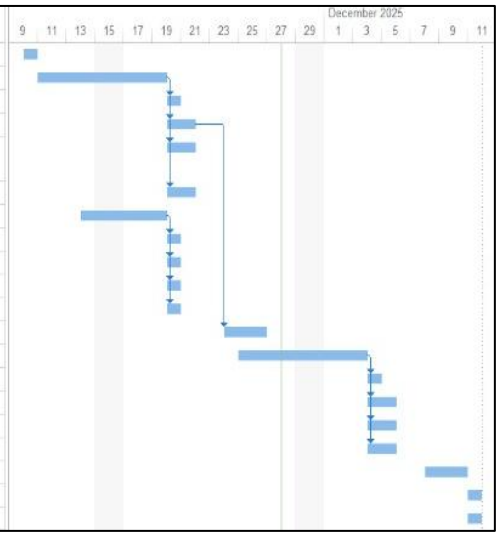

Figure 35: Gantt Chart.

8. Conclusion

In conclsion, the development of the Hotel Reservation Management System demonstrates the effectiveness of transforming traditional booking procedures into a fully automated digital workflow. Throughout the project, system requirements were analyzed, modeled, and implemented to support key hotel operations including room reservations, payment processing, staff management, promotions, reporting, and housekeeping control. The prototype confirms that the system can improve operational accuracy, increase efficiency, and provide a more convenient reservation experience for guests. By utilizing a secure layered architecture with role-based access, the system ensures data integrity and reliable performance during daily hotel activities. Although the system achieves its primary objectives, additional enhancements such as mobile application support, integration with external booking platforms, and intelligent analytics could further strengthen its usability and scalability. Overall, the HRMS presents a practical solution that reduces manual workload, minimizes reservation conflicts, and contributes to better hotel management and customer satisfaction.

9. References

[1] *Colcol, S. (2025, September 10). Hotel Reservation System: Software guide for 2025. SiteMinder.* https://www.siteminder.com/r/hotel-reservation-system/

[2] *Vaishnavi, S., Reddy, K. A., & Mahenderker, T. (2025, April). Online Hotel Reservation System. International Journal for Research in Applied Science & Engineering Technology (IJRASET), 13(4).* https://doi.org/10.22214/ijraset.2025.68464

[3] *Hu, P., & Gu, D. (2013). Development and Implementation of WEB-Based Online Hotel Reservation System. In Proceedings of the 2nd International Symposium on Computer, Communication, Control and Automation (ISCCCA-13). Atlantis Press.* https://doi.org/10.2991/isccca.2013.58

[4] *Ameh, J. J. (2016). A design and implementation of online hotel management system (Case study). Academia.edu.* https://www.academia.edu/36302522/A_DESIGN_AND_IMPLEMENTATION_OF_ONLINE_HOTEL_MANAGEMENT_SYSTEM_CASE_STUDY

[5] *Black Duck Editorial Staff. (2017, April 4). Attributes of secure web application architecture. Black Duck.* https://www.blackduck.com/blog/attributes-of-secure-web-application-architecture.html

[6] *Koh, W. S., & Hassim, Y. M. M. (2021). Hotel reservation management system. Applied Information Technology and Computer Science, 2(2), 973–992.* https://doi.org/10.30880/aitcs.2021.02.02.061

[7] *tecnovy. (2025, September). Top 10 software architecture & design patterns of 2025. Retrieved December 4, 2025, from* https://tecnovy.com/en/top-10-software-architecture-patterns *(*tecnovy.com*)*